Status	Finished
Started	Monday, 26 October 2026, 6:26 PM
Completed	Monday, 26 October 2026, 6:32 PM
Duration	6 mins 4 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By starting from the first element
b. By dividing the list length by two ✓
c. By using a hash function
d. By comparing each element sequentially

Question **2**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It requires the list to be sorted ✓
b. It does not work with negative numbers
c. It can only be applied to large lists
d. It is slower than linear search for small lists

Question **3**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

a. Hashing
b. Binary search
c. Jump search
d. Linear search ✓

Question **4**

Incorrect

Mark 0.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

- a. The middle element is chosen randomly
- b. The higher middle element is chosen as the middle element
- c. The search stops
- d. The lower middle element is chosen as the middle element ✗

Question **5**

Correct

Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

- a. Linear search ✓
- b. Hash search
- c. Interpolation search
- d. Binary search

Question **6**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

- a. When the item is somewhere in the middle of the array ✓
- b. Item is the last element in the array or item is not there at all
- c. When the item is not the array at all
- d. When the item is the last element in the array

Question **7**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

- a. Binary search ✓
- b. Linear search
- c. Hash search
- d. Sequential search

Question **8**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. The list must be sorted

b. There should be direct access to the middle element in any sublist

c. Number values should only be present

Number values should only be present

d. There must be a mechanism to delete and/or insert elements in the list ✓

Question **9**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

a. Dynamic search

b. Linear search ✓

c. Hashing

d. Binary search

Question **10**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search can be applied to both sorted and unsorted lists. ✓

b. Linear search divides the list into halves.

c. Linear search is more efficient than binary search on large lists.

d. Linear search requires the list to be sorted.

Question **11**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

- a. Checking each element sequentially ✓
- b. Dividing the list in half repeatedly
- c. Sorting the list before searching
- d. Skipping every second element

Question **12**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

- a. O(log n)
- b. O(n log n)
- c. O(1)
- d. O👎 ✓

Question **13**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

- a. Complexity
- b. Selection
- c. Insertion
- d. Bubble ✓

Question **14**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list contains duplicate elements
b. When the list is unsorted
c. When the list is very small
d. When the list is sorted ✓

Question **15**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. n ✓
b. log n
c. n/2
d. 1